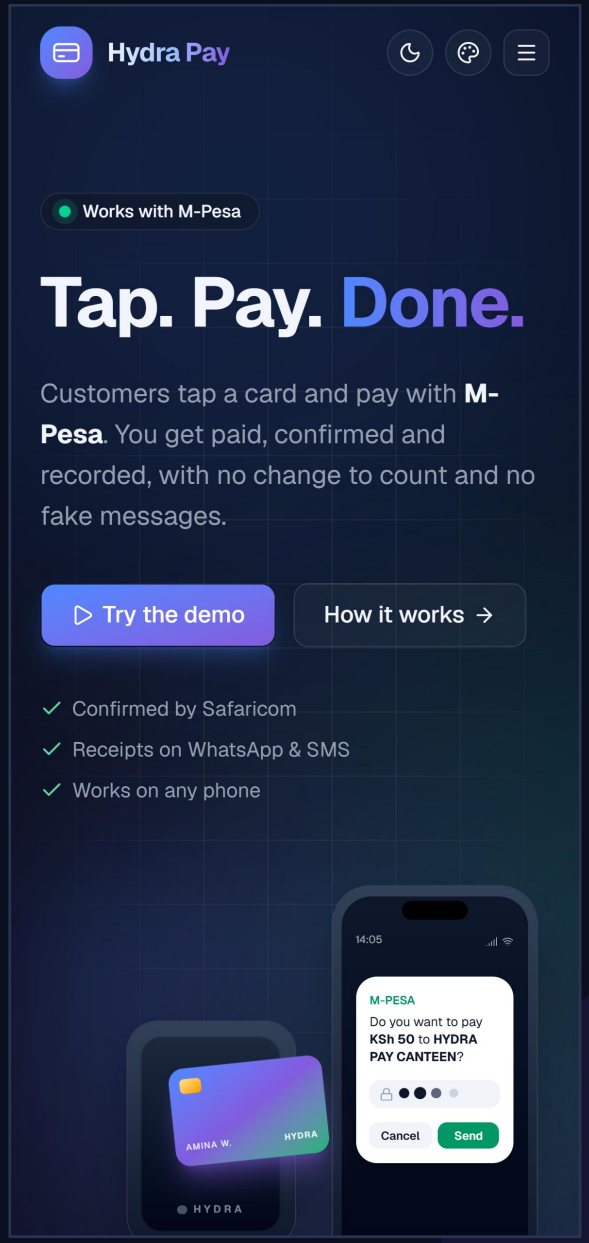
Works with M-Pesa

Tap. Pay. Done.

Hydra Pay is a low-cost tap-to-pay system for busy small businesses. Customers tap a card and pay with M-Pesa. Every payment is confirmed by Safaricom and recorded automatically.

Faster queues | Verified payments | Clear records

Team HYDRA-PAY
Technical University of Kenya
Live: hydra-pay.vercel.app



THE PROBLEM

Busy small businesses lose time and money at the till

1

Cash slows everything

Counting money and finding change at every sale builds queues. Cash leaves no record and is a target for theft.

2

Mobile money disputes

Fake confirmation SMSs, screenshots, reversals and "I've sent it" arguments. Checking each payment holds up the queue.

3

Records don't add up

Sales sit in a notebook, the till and a phone inbox. At the end of the day owners can't tell what they earned.

Owners need an affordable way to **accept, verify and record** payments without slowing down.

MARKET EVIDENCE

Cash and M-Pesa live side by side

99%

of surveyed small businesses run partly or wholly in cash

86%

accepted digital payments by June 2023

Both

mobile money is everywhere, but cash is still how many get paid

Sources: FinAccess MSE Tracker Survey 2023; World Bank analysis of Kenya's mobile money market.

Hydra Pay works with the M-Pesa habits people already have. **The opportunity is making every transaction faster and easier to track.**

OUR SOLUTION

One tap: pay, verify and record in one flow

1

Tap

The customer taps a card, key tag or wristband on the reader.

2

M-Pesa prompt

The price pops up on the customer's phone, paid to the business till.

3

Verify

They enter their PIN. Paid only when Safaricom confirms.

4

Record

Saved with amount, time and M-Pesa code. Receipt by WhatsApp or SMS.

Pay each time

Every tap sends one M-Pesa prompt for the price. Nothing to set up for customers.

Prepaid card

Regulars top up once with M-Pesa or cash. Each tap deducts instantly, with no PIN needed.

THE PRODUCT

A live dashboard for the owner, a website for customers



- **Live totals:** money collected, paid and unpaid taps, who is inside.
- **Live feed:** every tap and payment the moment it happens.
- **Card management:** register, top up, block or move a lost card.
- **Works on any screen:** phone, tablet or desktop, light or dark, six colour themes.
- **Exports:** daily records to Excel (CSV).
- **Public website:** hydra-pay.vercel.app with a try-it-yourself demo.

HOW IT'S BUILT

Simple, low-cost hardware with verified money behind it



WhatsApp
receipts, bot,
AI in 3 languages

SMS
receipts and alerts
via ThriveVAS

USSD
menu for any phone,
no smartphone needed

Records
cards, taps and payments,
nothing deleted

- **Verified money:** paid only on Safaricom's answer, recorded exactly once, even if the confirmation arrives twice.
- **Reliable reads:** the reader filters double taps and retries half-read cards; it reconnects by itself if unplugged.
- **Auditable:** an undo is a new reversal entry, so the full history is always kept.
- **Protected:** every change, including sending a payment prompt, needs the owner's admin PIN.

PROOF OF WORK

It works today, and you can try it

27
M-Pesa prompts sent

10
confirmed paid by Safaricom

15
declined or unanswered, correctly NOT paid

Daraja sandbox test runs, 22–23 October 2026.

Live demo at our table

Tap a card: watch the M-Pesa prompt arrive, the dashboard update and the receipt land.

Working today

- ✓ Card and key-tag taps logged as IN / OUT
- ✓ M-Pesa prompt to the customer's phone for the price
- ✓ Receipts with the M-Pesa code on WhatsApp and SMS
- ✓ Prepaid wallet with M-Pesa or cash top-ups
- ✓ WhatsApp bot and AI help in English, Kiswahili and Sheng
- ✓ Live dashboard on phone and desktop, with CSV export
- ✓ Lost card: move the account to a new card in one click
- ✓ USSD menu for phones without internet
- ✓ Public website and demo: hydra-pay.vercel.app

TARGET MARKET

Businesses where every second counts

PILOT FIRST

Canteens & cafés

Lunch-hour rush, fixed prices, regular customers.

Need: shorter checkout and accurate sales records.

PILOT FIRST

Shops & market stalls

Mixed cash and M-Pesa every day.

Need: easy payment tracking without expensive software.

NEXT

Matatus & transport

Many small fares under time pressure. Tap IN to board.

Need: fast fare collection and clear daily takings.

Customer: the business owner or operator. **Users:** the person paying and the person receiving. We start where we can observe and measure, then move to matatus.

WHY HYDRA PAY

Better than cash, a till number or a notebook

	Hydra Pay	Cash	Till number	Notebook	Card machine
No change, nothing to type	✓	✗	~	✗	~
Payment checked with Safaricom	✓	✗	~	✗	✓
Every sale saved automatically	✓	✗	~	✗	✓
Daily totals ready	✓	✗	~	~	~
Receipt for the customer	✓	✗	✓	✗	✓
Works without a smartphone	✓	✓	✓	✓	~
No cut taken from each sale	✓	✓	~	✓	✗

✓ yes ~ partly ✗ no · Standard M-Pesa charges still apply.

BUSINESS MODEL

Affordable for the seller, sustainable for us

1

Freemium

Basic transaction recording free; advanced reports and features paid.

2

Subscription

An affordable weekly or monthly fee per reader.

3

Business packages

Fleet and multi-stall dashboards for SACCOs and larger vendors.

4

Partnerships

Matatu SACCOs, market managers, canteen operators and business associations.

No cut of each sale: low-value, high-volume sellers keep their margin. We'll test willingness to pay in the pilot before fixing prices.

FEASIBILITY & SECURITY

Risks we've designed for

Challenge	How Hydra Pay handles it
Fake confirmations	Status comes only from Safaricom's Daraja API. An unconfirmed payment is never shown as paid.
Reversals & disputes	Every payment keeps its M-Pesa code, status and time in a history that is never deleted.
Slow checkout	Tap, then PIN. No typing. Checkout time will be measured in the pilot.
Poor network	Pending is kept separate from paid and never marked failed too early. Offline tap queue is next.
Tight budgets	Off-the-shelf ESP32 + RC522, no POS terminal, no per-sale fee.
Misuse & data	Admin PIN for every change; keys kept out of code; minimal data, aligned with Kenya's Data Protection Act.

ROADMAP & VALIDATION

From working prototype to real businesses

DONE

Prototype

Reader, M-Pesa, verification, records, WhatsApp, SMS and USSD. Dashboard and website online.

NEXT

Discovery

10–20 interviews with vendors and conductors about payments, disputes and records.

NEXT

Go live

Live Safaricom till, live USSD short code, offline reader that syncs later.

PILOT

Controlled pilot

A few canteens and traders, measured against a before baseline.

LATER

Expand

Matatu SACCOs, then more payment options. The owner always gets KES.

We'll measure: transaction time · queue time at peak · % recorded correctly · verification success · reconciliation errors · merchant satisfaction and willingness to pay. Targets are set after the baseline.

INDUSTRY ENGAGEMENT

Meeting Safaricom leadership



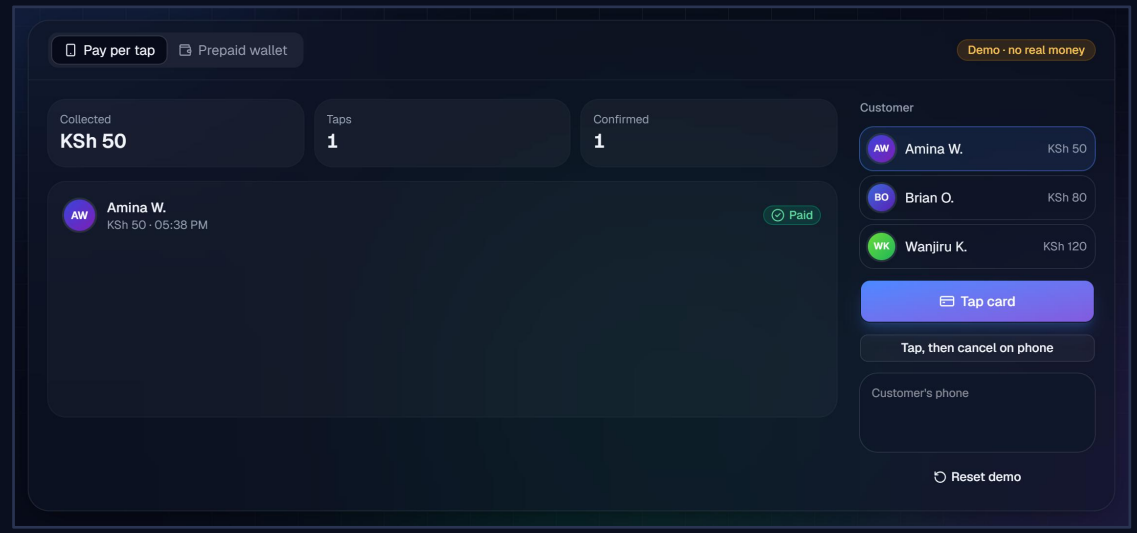
With Safaricom's former Chief Financial Services Officer



With Safaricom's Chief Technology & Information Officer

Tap.
Pay.
Done.

Fast payments, safer transactions, clearer records.



The try-it demo on our website

Scan to try the demo

hydra-pay.vercel.app

Questions? Come tap a real card at our table.